

ADV: Logs and Exponentials (Adv), E1 Logs and Exponentials (Adv)
Exponential Calculus (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 64.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

SUMMARY REVISION

- This subtopic is a database of *Exponential Calculus* that covers exponential differentiation and integration questions in the one location.
- It is important to note that this sub-topic is a collection of all questions within *Calculus C2 and C4* that involve exponentials, providing a convenient location for comprehensively revising exponential calculus.

HSC ANALYSIS - What to expect and common pitfalls

- *Exponential Calculus* (2.1%) has appeared regularly in dedicated questions in past Advanced exams, most recently in 2020 where it was examined twice. It represents a great opportunity for high scoring.
- Differentiating and then integrating surprisingly caused problems in the 2020 exam and deserves attention (see 2020 Adv 18 and 2016 Adv 12d).
- Students are strongly advised to review the differentiation of $a^{f(x)}$ which receives its own dot point in the new syllabus - questions C2 SM-Bank 2, 3 and 4 provide coverage of this area. This has appeared in significant cross-topic questions in both new syllabus exams (covered in the later topics).

Questions

1. Calculus, 2ADV C4 2014 HSC 4 MC

Which expression is equal to $\int e^{2x} dx$?

- (A) $e^{2x} + C$
(B) $2e^{2x} + C$
(C) $\frac{e^{2x}}{2} + C$
(D) $\frac{e^{2x+1}}{2x+1} + C$

2. Calculus, 2ADV C2 2017 HSC 3 MC

What is the derivative of e^{x^2} ?

- (A) $x^2 e^{x^2-1}$
(B) $2xe^{2x}$
(C) $2xe^{x^2}$
(D) $2e^{x^2}$

3. Calculus, 2ADV C4 2020 HSC 4 MC

What is $\int e + e^{3x} dx$?

- A. $ex + 3e^{3x} + c$
B. $ex + \frac{1}{3}e^{3x} + c$
C. $e + 3e^{3x} + c$
D. $e + \frac{1}{3}e^{3x} + c$

4. Calculus, 2ADV C2 SM-Bank 7

Let $f(x) = \frac{e^x}{(x^2 - 3)}$.

Find $f'(x)$. (2 marks)

5. Calculus, 2ADV C2 2019 MET1 1a

Let $y = \frac{2e^{2x} - 1}{e^x}$.

Find $\frac{dy}{dx}$. (2 marks)

6. Calculus, 2ADV C2 2009 HSC 2aii

Differentiate with respect to x .

$(e^x + 1)^2$. (2 marks)

7. Calculus, 2ADV C2 2012 HSC 11d

Differentiate $(3 + e^{2x})^5$. (2 marks)

8. Calculus, 2ADV C2 2013 HSC 11d

Differentiate $x^2 e^x$ (2 marks)

9. Calculus, 2ADV C4 2013 HSC 11e

Find $\int e^{4x+1} dx$ (2 marks)

10. Calculus, 2ADV C4 2004 HSC 3bi

Evaluate $\int_1^2 e^{3x} dx$. (2 marks)

11. Calculus, 2ADV C4 2006 HSC 2bi

Find $\int 1 + e^{7x} dx$. (2 marks)

12. Calculus, 2ADV C2 2007 HSC 2ai

Differentiate with respect to x :

$\frac{2x}{e^x + 1}$ (2 marks)

13. Calculus, 2ADV C2 2015 HSC 11e

Differentiate $(e^x + x)^5$. (2 marks)

14. Calculus, 2ADV C4 2018 HSC 11e

Evaluate $\int_0^3 e^{5x} dx$. (2 marks)

15. Calculus, 2ADV C2 2018 HSC 11g

Differentiate $\frac{e^x}{x+1}$. (2 marks)

16. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :

$e^{\tan(2x)}$ (2 marks)

17. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :

10^{5x^2-3x} . (2 marks)

18. Calculus, 2ADV C2 EQ-Bank 3

Differentiate $3x 6^x$. (2 marks)

19. Calculus, 2ADV C4 2016 HSC 12d

i. Differentiate $y = xe^{3x}$. (1 mark)

ii. Hence find the exact value of $\int_0^2 e^{3x}(3+9x) dx$. (2 marks)

20. Calculus, 2ADV C2 SM-Bank 4

Differentiate $5^{x^2} 5x$. (2 marks)

21. Calculus, 2ADV C4 2019 MET1-N 3

Evaluate $\int_0^1 e^x - e^{-x} dx$. (2 marks)

22. Calculus, 2ADV C4 2020 HSC 18

a. Differentiate $e^{2x}(2x+1)$. (2 marks)

b. Hence, find $\int (x+1)e^{2x} dx$. (1 marks)

Worked Solutions

1. Calculus, 2ADV C4 2014 HSC 4 MC

$$\begin{aligned}\int e^{2x} dx \\&= \frac{1}{2}e^{2x} + C \\&\Rightarrow C\end{aligned}$$

2. Calculus, 2ADV C2 2017 HSC 3 MC

$$\begin{aligned}y &= e^{x^2} \\ \frac{dy}{dx} &= 2x e^{x^2} \\ &\Rightarrow C\end{aligned}$$

3. Calculus, 2ADV C4 2020 HSC 4 MC

$$\begin{aligned}\int e + e^{3x} &= ex + \frac{1}{3}e^{3x} + c \\ \text{(Note } e \text{ is a simple constant here)} \\ &\Rightarrow B\end{aligned}$$

4. Calculus, 2ADV C2 SM-Bank 7

$$\begin{aligned}\text{Let } u &= e^x \Rightarrow u' = e^x \\ v &= (x^2 - 3) \Rightarrow v' = 2x \\ f'(x) &= \frac{e^x(x^2 - 3) - 2xe^x}{(x^2 - 3)^2} \\ &= \frac{e^x(x^2 - 2x - 3)}{(x^2 - 3)^2}\end{aligned}$$

5. Calculus, 2ADV C2 2019 MET1 1a

Method 1

$$y = 2e^x - e^{-x}$$

$$\frac{dy}{dx} = 2e^x + e^{-x}$$

Method 2

$$\frac{dy}{dx} = \frac{4e^{2x} \cdot e^x - (2e^{2x} - 1)e^x}{(e^x)^2}$$

$$= \frac{4e^{3x} - 2e^{3x} + e^x}{e^{2x}}$$

$$= \frac{2e^{2x} + 1}{e^x}$$

6. Calculus, 2ADV C2 2009 HSC 2aii

$$y = (e^x + 1)^2$$

$$\begin{aligned}\frac{dy}{dx} &= 2(e^x + 1)^1 \times \frac{d}{dx}(e^x + 1) \\ &= 2e^x(e^x + 1)\end{aligned}$$

7. Calculus, 2ADV C2 2012 HSC 11d

$$y = (3 + e^{2x})^5$$

$$\frac{dy}{dx} = 5(3 + e^{2x})^4 \times \frac{d}{dx}(3 + e^{2x})$$

$$= 5(3 + e^{2x})^4 \times 2e^{2x}$$

$$= 10e^{2x}(3 + e^{2x})^4$$

8. Calculus, 2ADV C2 2013 HSC 11d

Using the product rule

$$\text{Let } u = x^2, \quad u' = 2x$$

$$\text{Let } v = e^x, \quad v' = e^x$$

$$\frac{d(uv)}{dx} = u'v + v'u$$

$$= 2xe^x + x^2e^x$$

$$= xe^x(x + 2)$$

9. Calculus, 2ADV C4 2013 HSC 11e

$$\int e^{4x+1} dx = \frac{1}{4}e^{4x+1} + C$$

10. Calculus, 2ADV C4 2004 HSC 3bi

$$\int_1^2 e^{3x} dx = \frac{1}{3}[e^{3x}]_1^2$$

$$= \frac{1}{3}(e^6 - e^3)$$

11. Calculus, 2ADV C4 2006 HSC 2bi

$$\int 1 + e^{7x} dx = x + \frac{1}{7}e^{7x} + c$$

12. Calculus, 2ADV C2 2007 HSC 2ai

$$\begin{aligned}
 y &= \frac{2x}{e^x + 1} \\
 u &= 2x \quad v = e^x + 1 \\
 u' &= 2 \quad v' = e^x \\
 \frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\
 &= \frac{2(e^x + 1) - 2x(e^x)}{(e^x + 1)^2} \\
 &= \frac{2e^x + 2 - 2x \cdot e^x}{(e^x + 1)^2} \\
 &= \frac{2(e^x + 1 - xe^x)}{(e^x + 1)^2}
 \end{aligned}$$

13. Calculus, 2ADV C2 2015 HSC 11e

$$\begin{aligned}
 y &= (e^x + x)^5 \\
 \frac{dy}{dx} &= 5(e^x + x)^4 \times \frac{d}{dx}(e^x + x) \\
 &= 5(e^x + x)^4 \times (e^x + 1) \\
 &= 5(e^x + 1)(e^x + x)^4
 \end{aligned}$$

14. Calculus, 2ADV C4 2018 HSC 11e

$$\begin{aligned}
 \int_0^3 e^{5x} dx &= \left[\frac{1}{5} e^{5x} \right]_0^3 \\
 &= \frac{1}{5} e^{15} - \frac{1}{5} e^0 \\
 &= \frac{1}{5} (e^{15} - 1)
 \end{aligned}$$

15. Calculus, 2ADV C2 2018 HSC 11g

$$y = \frac{e^x}{x + 1}$$

Differentiate using quotient rule:

$$u = e^x \quad v = x + 1$$

$$u' = e^x \quad v' = 1$$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\
 &= \frac{e^x(x + 1) - e^x \cdot 1}{(x + 1)^2} \\
 &= \frac{xe^x}{(x + 1)^2}
 \end{aligned}$$

16. Calculus, 2ADV C2 EQ-Bank 2

$$\begin{aligned}
 y &= e^{\tan(2x)} \\
 \frac{dy}{dx} &= \frac{d}{dx} \tan(2x) \times e^{\tan(2x)} \\
 &= 2 \sec^2(2x) \cdot e^{\tan(2x)}
 \end{aligned}$$

17. Calculus, 2ADV C2 EQ-Bank 2

$$y = 10^{5x^2 - 3x}$$

$$\frac{dy}{dx} = \ln 10 (10x - 3) \cdot 10^{5x^2 - 3x}$$

TIP: The new Advanced reference sheet can be used here!

18. Calculus, 2ADV C2 EQ-Bank 3

$$y = 3x \cdot 6^x$$

$$\begin{aligned}
 \frac{dy}{dx} &= 3 \cdot 6^x + \ln 6 \cdot 6^x \cdot 3x \\
 &= 3 \cdot 6^x (1 + x \ln 6)
 \end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 6^x .

19. Calculus, 2ADV C4 2016 HSC 12d

i. $y = xe^{3x}$

Using product rule:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot 3e^{3x} + 1 \cdot e^{3x} \\ &= e^{3x}(1 + 3x)\end{aligned}$$

$$\begin{aligned}\text{ii. } \int_0^2 e^{3x}(3 + 9x) dx &= 3 \int_0^2 e^{3x}(1 + 3x) dx \\ &= 3[xe^{3x}]_0^2 \\ &= 3(2e^6 - 0) \\ &= 6e^6\end{aligned}$$

20. Calculus, 2ADV C2 SM-Bank 4

$y = 5^{x^2} \cdot 5x$

$$\begin{aligned}\frac{dy}{dx} &= \ln 5 \cdot 2x \cdot 5^{x^2} \cdot 5x + 5^{x^2} \cdot 5 \\ &= 5^{x^2}(\ln 5 \cdot 10x^2 + 5) \\ &= 5^{x^2+1}(\ln 5 \cdot 2x^2 + 1)\end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 5^x .

21. Calculus, 2ADV C4 2019 MET1-N 3

$$\begin{aligned}\int_0^1 e^x - e^{-x} dx &= [e^x + e^{-x}]_0^1 \\ &= \left[e + \frac{1}{e} - (1 + 1)\right] \\ &= e + \frac{1}{e} - 2\end{aligned}$$

22. Calculus, 2ADV C4 2020 HSC 18

a. $y = e^{2x}(2x + 1)$

$$\begin{aligned}\frac{dy}{dx} &= 2x^{2x}(2x + 1) + 2e^{2x} \\ &= 2e^{2x}(2x + 2) \\ &= 4e^{2x}(x + 1)\end{aligned}$$

♦ Mean mark part (b) 40%.

$$\begin{aligned}\text{b. } \int (x + 1)e^{2x} dx &= \frac{1}{4} \int 4e^{2x}(x + 1) \\ &= \frac{1}{4}e^{2x}(2x + 1) + c\end{aligned}$$